# Customer Analytics (MKTG 562) — Autumn 2025

University of Washington | Foster School of Business

## Course Information

**Instructor:** Reza Roshangarzadeh
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**Additional Office Hours:** By appointment

**Section A/B:** Tue & Thu 13:30 am – 15:20 pm (Paccar 393)
**Lab Session:** TBA

## Course Objectives

Customer Analytics explores how businesses can leverage data analytics to understand and market to individual customers effectively. In today's data-driven environment, marketing is transforming from art into a science. Many companies collect extensive information about consumer behaviors and reactions to marketing campaigns but lack the expertise to derive actionable insights. This course bridges that gap, teaching students how to approach marketing scientifically through the application of data analytics to customer-level data along with managerial recommendations.

Students will gain hands-on experience with tools such as data, and analytics to collect, analyze, and act on customer information. While quantitative methods are central to the course, the primary goal is to develop competency in managing and interacting with marketing analytics teams rather than mastering advanced statistics or data science. This foundational skill set enables students to take advantage of the potential of Big Data for impactful decision-making within the realm of marketing.

## Course Format

This course combines lectures, readings, and applied assignments to guide students through the full lifecycle of customer analytics. We begin by learning how to identify marketing problems that can be addressed with data, then move to choosing the right analytical approach for each situation. Students will work directly in Python on real-world datasets, gaining hands-on experience not just in running analyses but also in turning the results into clear, actionable marketing strategies.

The emphasis throughout the course is on practical, job-ready skills. You will learn how to use individual-level customer data to segment and target customers more effectively, recommend products tailored to their needs, anticipate purchasing patterns, and design strategies to retain valuable customers. By the end of the course, you will be equipped with tools that can be put to use immediately in a professional setting.

For those who want to go further, the Foster Marketing Analytics Specialization also offers complementary courses such as Analytics for Marketing Decisions, Digital Marketing Analytics, and Pricing Strategy and Analytics. These courses explore topics such as automated product development, algorithmic price optimization, online advertising, platform design, and social media analytics.

## Course Organization

This course is structured around several key themes that reflect the core challenges faced by marketing analysts and decision-makers:

**Understanding customer value:** We begin by exploring why customer-level data is central to modern marketing strategy. This section highlights how firms measure customer value, why some customers are more profitable than others, and how advances in data analytics have changed the way firms build long-term customer relationships.

**Prospecting and targeting the right customers:** In this section, we focus on customer acquisition. We discuss how to identify promising prospects, decide which customers are most worthwhile to target, and design individualized offers that maximize conversion potential.

**Developing customers:** This section emphasizes strategies for improving customer profitability over time. Topics include upselling and cross-selling, recommending new products, and using data (such as reviews and ratings) to inform personalized offers.

**Retaining customers:** Retention is one of the most important levers for sustainable growth. We will cover methods to predict which customers are at risk of leaving, strategies for reducing churn, and how to design incentives that encourage loyalty.

**Pitfalls and challenges in customer analytics:** While analytics can provide powerful insights, it also brings risks if not applied carefully. In this section, we discuss common mistakes, misinterpretations, and ethical issues that can arise when working with customer data, as well as best practices for using analytics responsibly.

## Grading Policy and Assignments

| | |
|---|---|
| **Individual homework assignments (2 × 14% each)** | **28%** |
| **Group homework assignments (3 × 18% each)** | **54%** |
| **Individual reflections (4 × 1.5% each)** | **6%** |
| **Group contribution and class participation** | **12%** |

Grades will be curved at the end of the quarter to satisfy an appropriate grade distribution.

### Assignments

Assignments are a central part of this course. They are designed not only to give you practice with the tools and concepts we cover but also to help you develop confidence in applying analytics to real-world business problems. Much of the learning in this class will come from these assignments, and they will balance both individual and group work.

Individual assignments must be completed on your own. Collaboration with other students is not permitted for these, and your write-up should reflect only your own work. Group assignments will be completed in self-selected groups of 2–4 students. Only one write-up per group should be submitted. Groups should be formed at the beginning of the quarter and remain consistent throughout the course. The goal of these assignments is to help you practice two key skills: First, Analyzing customer data using basic coding and statistical methods, and second, communicating insights in a way that makes sense to a managerial audience.

You do not need to arrive with deep programming or statistics experience. The assignments will not assume advanced technical backgrounds, and any tools or methods you need will be explained as we go. Our focus is on using analytics to answer meaningful business questions and to make clear, persuasive recommendations.

By the end of the course, you should feel comfortable moving from data to insight, and from insight to action. The assignments are designed to give you that experience step by step. Each assignment will be discussed extensively in the class on the day it is due.

### Individual Reflections

During the course I will ask you to submit short reflections (max 1/2 page) about what you learned from a lecture and/or an assignment. These reflections are always individual work, even if the assignment you are asked to discuss was a group assignment.

### Class Participation and Group Contributions

Active participation is a key part of the learning experience in this course. Because many of the cases and assignments we will discuss do not have one single "right" answer, the goal of participation is not to be perfect, but to be engaged, open-minded, and willing to share your perspective. High-quality contributions are those that connect to the topic at hand, move the discussion forward, or build thoughtfully on what others have said. You are encouraged to take risks, ask questions, and explore ideas during class discussions. What matters most is not always arriving at the correct solution, but showing curiosity, effort, and a willingness to contribute to our collective learning.

Each student should submit a confidential peer evaluation form at the end of quarter. This helps me understand group dynamics when you are working on group assignments together so that I will be able to differentiate between responsible group members and those who might not have done their assigned parts in the group. This ensures that the effort and input of each group member are recognized fairly. The responses here greatly affect student's score on the group contribution and class participation component of grading.

## Course Materials

Readings in this course consist of a mixture of slides and additional readings. No textbook is required.

**Textbook:** None required
**Course pack:** None required
**Lecture videos:** Distributed on Canvas
**Lecture notes:** Distributed on Canvas
**Articles and additional course material:** Distributed on Canvas

The syllabus, lecture videos, lecture notes, and additional course material are all posted on Canvas. All material is organized by the class number it belongs to.

## Software

In this course we will be using Python for data analysis, with all coding done through Google Colab. Colab is a cloud-based platform, which means you do not need to install any special programs on your computer. Everything runs directly on your web browser, and Google provides the computing resources. This makes it accessible from virtually any laptop, regardless of operating system or processing power.

Python is the most widely used programming language in business and data science today, and gaining familiarity with it will be a valuable takeaway from the course. At the same time, you do not need any prior experience with Python to succeed here. We will learn everything step by step, and I will provide support in class to help you get comfortable with the basics.

If you encounter technical issues or need help with coding, I can assist with troubleshooting. In addition, Python has an extensive online community and countless free resources, so you will always be able to find guidance as you continue practicing beyond this course.

## Honor Code

Students are expected to respect the Foster code of conduct at all times.

Written assignments will be either individual assignments or group assignments, as specified on Canvas. For individual assignments, you are not allowed to work with other students – the write-up should reflect your own work alone.

It will be a violation of academic integrity if you base any of your assignments on discussions or solutions that you find online or that are shared by people who have taken similar courses in the past. I reserve the right to fail you for the course if I become aware of such a violation. To ensure that future classes benefit from a similar learning experience, do not share any case or assignment materials with students outside the class.

## Religious Accommodations

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at Religious Accommodations Policy.

Accommodations must be requested within the first two weeks of this course using the Religious Accommodations Request form.

## Access and Accommodations

Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but are not limited to mental health, attention-related, learning, vision, hearing, physical or health impacts), you are welcome to contact DRS at 206-543-8924 or uwdrs@uw.edu or disability.uw.edu. DRS offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions. Reasonable accommodations are established through an interactive process between you, your instructor(s) and DRS. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law.

## Tentative Schedule

| Date | # | Session Topics | Assignment Due |
|---|---|---|---|
| Sep 25 | 1 | Course overview | |
| Sep 30 | 2 | Using Python for customer analytics<br>Customer Lifetime Value (LTV) | |
| Oct 2 | 3 | Doing correct analysis | |
| Oct 7 | 4 | Causality and experimentation | |
| Oct 9 | 5 | Case Analysis: Quip<br>Customer response prediction with RFM | Quip |
| Oct 14 | 6 | Statistics overview | |
| Oct 16 | 7 | Customer response prediction with Logit | |
| Oct 21 | 8 | Case Analysis: DoorDash<br>Lift and Gains graphs | DoorDash |
| Oct 23 | 9 | Interaction effects<br>Customer response prediction with Neural Networks | |
| Oct 28 | 10 | Model performance | |
| Oct 30 | 11 | Customer response prediction with Decision Trees | |
| Nov 4 | 12 | Case Analysis: Stumptown | Stumptown |
| Nov 6 | 13 | Next-Product-To-Buy models | *Reflection* on Logit |
| Nov 13 | 14 | Case Analysis: Nuova Simonelli | Nuova Simonelli |
| Nov 18 | 15 | Recommendation systems | *Reflection* on Nuova Simonelli |
| Nov 20 | 16 | Predicting and managing churn | |
| Nov 25 | 17 | Case Analysis: IKEA | IKEA |
| Dec 2 | 18 | Pitfalls of customer analytics | *Reflection* on NPTB models |
| Dec 4 | 19 | Course wrap-up | *Reflection* on course |

Understanding Customer Value
Getting Ready for Analytics
Prospecting & Targeting
Developing Customers
Retaining Customers
Course Wrap-up

(End of syllabus)